

Adaptive trade-offs in juvenile salmonid metabolism associated with habitat partitioning between coho salmon and steelhead trout in coastal streams

Travis E. Van Leeuwen^{*,1}, Jordan S. Rosenfeld² and Jeffrey G. Richards¹

¹Department of Zoology, University of British Columbia, Vancouver, BC V6T 1Z4, Canada; and ²British Columbia Ministry of the Environment, 2202 Main Mall, University of British Columbia, Vancouver, BC V6T 1Z4, Canada

Summary

1. Adaptive trade-offs are fundamental to the evolution of diversity and the coexistence of similar taxa and occur when complimentary combinations of traits maximize efficiency of resource exploitation or survival at different points on environmental gradients.

2. Standard metabolic rate (SMR) is a key physiological trait that reflects adaptations to baseline metabolic performance, whereas active metabolism reflects adaptations to variable metabolic output associated with performance related to foraging, predator avoidance, aggressive interactions or migratory movements. Benefits of high SMR and active metabolism may change along a resource (productivity) gradient, indicating that a trade-off exists among active metabolism, resting metabolism and energy intake.

3. We measured and compared SMR, maximal metabolic rate (MMR), aerobic scope (AS), swim performance (UCrit) and growth of juvenile hatchery and wild steelhead and coho salmon held on high- and low-food rations in order to better understand the potential significance of variation in SMR to growth, differentiation between species, and patterns of habitat use along a productivity gradient.

4. We found that differences in SMR, MMR, AS, swim performance and growth rate between steelhead trout and coho salmon were reduced in hatchery-reared fish compared with wild fish. Wild steelhead had a higher MMR, AS, swim performance and growth rate than wild coho, but adaptations between species do not appear to involve differences in SMR or to trade-off increased growth rate against lower swim performance, as commonly observed for high-growth strains. Instead, we hypothesize that wild steelhead may be trading off higher growth rate for lower food consumption efficiency, similar to strategies adopted by anadromous vs. resident brook trout and Atlantic salmon vs. brook trout. This highlights potential differences in food consumption and digestion strategies as cryptic adaptations ecologically differentiating salmonid species.

5. We hypothesize that divergent digestive strategies, which are common and well documented among terrestrial vertebrates, may be an important but overlooked aspect of adaptive strategies of juvenile salmonids, and fish in general.

Key-words: adaptive trade-offs, food ration, habitat partitioning, hatchery vs. wild, life-history, *Oncorhynchus kisutch*, *Oncorhynchus mykiss*, salmonid growth

Introduction

Adaptive trade-offs are fundamental to the evolution of diversity and the coexistence of similar taxa (Schluter 1995) and occur when complimentary combinations of traits maximize efficiency of resource exploitation or survival at different points along environmental gradients (e.g. Biro & Stamps

2008; Careau *et al.* 2010; Finstad *et al.* 2011). For example, morphological adaptation to feeding on benthic vs. planktonic resources (gill raker spacing, body morphology, mouth orientation, etc.) is a common adaptive trade-off among freshwater fishes (Schluter 1993; Robinson & Wilson 1994) and matches habitat and resource use along a benthic–pelagic gradient.

In contrast with morphological traits, physiological adaptations are usually cryptic and are consequently more poorly documented, but likely as important in ecological

*Correspondence author. E-mail: t.vanLeeuwen@fisheries.ubc.ca
Order or authors represents their contribution to the study.

differentiation (Garland & Adolph 1991; Djawadan, Rose & Bradley 1997; Lee *et al.* 2003; Bochdansky *et al.* 2005; Killen *et al.* 2007). Although some physiological adaptations represent direct responses to simple selection gradients (e.g. hypoxia tolerance), others are the result of selection on complex life-history traits like growth or cost of transport. Standard metabolic rate (SMR) is a key physiological attribute that represents the integration of a broad suite of metabolic pathways and processes and is known to vary widely in animals (Boily 2002; Cutts, Metcalfe & Taylor 2002; Speakman, Krol & Johnson 2004). Because SMR is often positively correlated with growth (Metcalfe, Taylor & Thorpe 1995; Alvarez & Nicieza 2005) and constitutes a significant component of an animal's energy budget (Finstad *et al.* 2007), variation in SMR has been implicated in adaptive trade-offs between individuals (Biro & Stamps 2010), species (Steyermark 2002), ecotypes (Alvarez, Cano & Nicieza 2006; Ridgway 2008) and populations (Garland & Adolph 1991) of animals along productivity gradients (Mueller & Diamond 2001; Bozinovic *et al.* 2009). A high SMR may facilitate growth when resources are abundant, but elevated metabolic demands could result in decreased growth when resources are scarce (Forseth *et al.* 1999; Mueller & Diamond 2001; Ridgway 2008; Biro & Stamps 2010), suggesting that higher metabolic rates should be selected for in environments where resources are abundant (Steyermark 2002; Lovegrove 2003; Bochdansky *et al.* 2005). Although recent studies have demonstrated that the level of maintenance metabolism may affect foraging strategies and survival in birds and mammals (e.g. Mathot *et al.* 2009; Larivee *et al.* 2010), the role of SMR in adaptive trade-offs related to growth along productivity gradients remains unclear.

Maximizing energy intake (consumption) while minimizing energy expenditures (costs of foraging and transport) is an additional trade-off confronting organisms. While SMR integrates adaptations related to maintenance metabolism, the costs of foraging and predator avoidance represent active metabolism, and the optimal trade-off among active metabolism, resting metabolism and energy intake may also change along a resource (productivity) gradient. For instance, taxa adapted to abundant resources may be energy maximizers with higher growth, food consumption, and transport costs, while taxa adapted to habitat with less-abundant resources should be cost minimizers, with lower growth, energy intake, and costs of energy acquisition (Arendt 1997; Arendt & Wilson 1997; Tucker & Rasmussen 1999; Finstad *et al.* 2011). For terrestrial mammals, Lovegrove (2003) considered slow vs. fast metabolic adaptations to be one of the primary evolutionary responses to rainfall/productivity clines in terrestrial environments.

Juvenile steelhead trout (*Oncorhynchus mykiss*) and coho salmon (*Oncorhynchus kisutch*) occur in sympatry in many coastal streams in the Pacific Northwest of North America and represent model species for exploring the relationships between growth, metabolism, and performance physiology of fish adopting energy-maximizing vs. cost-minimizing strategies. Both species feed on aquatic and terrestrial inverte-

brates drifting in the water column, but partition habitat differently in the wild when found in sympatry. Coho forage near the surface (Fraser 1969; Johnston 1970) and occupy pools (Hartmann 1965; Bisson, Sullivan & Nielsen 1988; Bugert & Bjornn 1991), while steelhead hold closer to the substrate (Fraser 1969; Johnston 1970) and occupy faster velocity riffle habitat (Hartmann 1965; Bisson, Sullivan & Nielsen 1988; Bugert & Bjornn 1991). However, both species prefer and grow faster in deep, low-velocity pools (Hartmann 1965; Quinn & Peterson 1996; Young 2001) where they can feed on drift entering from the upstream riffle with minimal swimming costs. In many of these freshwater systems, coho typically emerge earlier than steelhead and therefore take up residence in preferred low-velocity pools (Young 2004), where their larger size and prior residence allow them to effectively displace steelhead into riffles (Hartmann 1965; Bisson, Sullivan & Nielsen 1988; Bugert & Bjornn 1991; Young 2004), leading to habitat partitioning. While pools provide optimal growth for both species, steelhead only realize higher growth rates in pools in the absence of larger competitively dominant coho, and steelhead appear to have evolved adaptations that allow them to also exploit riffle habitat where both swimming costs and productivity [in terms of prey encounter rates (Hayes, Stark & Shearer 2000)] are elevated.

In addition to partitioning habitat along a velocity and depth gradient, coho and steelhead differ morphologically in a manner consistent with their habitat use. Coho salmon have a laterally compressed body form with long median fins that facilitate rapid acceleration and turning, which would be advantageous for capturing invertebrate drift at the surface of pools (Bisson, Sullivan & Nielsen 1988). Steelhead, however, have a longer more cylindrical body form with shorter median fins that are best suited for holding in faster velocity riffle habitat as it minimizes drag (Bisson, Sullivan & Nielsen 1988). Previous studies also indicate that steelhead have higher maximum growth and food consumption than coho, which is consistent with maximizing energy intake at high foraging costs (Hartmann 1965; Sullivan *et al.* 2001; Young 2001), whereas coho have lower maximum growth and food consumption in pool habitats where costs of swimming are reduced (Sullivan *et al.* 2001; Young 2001).

Adaptive trade-offs along a productivity gradient also provide a framework for understanding the effect of hatcheries on physiological attributes that affect growth and survival. To better understand the effects of human selection on growth and metabolic trade-offs in these species, we compared wild steelhead and coho (McNab Creek, with no history of hatchery influence) to a partially domesticated stock of steelhead and coho (Chapman Creek, where a hatchery has been operating for over 20 years). Because hatcheries represent a resource-rich environment that selects for higher juvenile growth rates (Gross 1998), we hypothesized that potential differences in growth, metabolism and swimming performance between juvenile steelhead and coho would be reduced (relative to wild stocks) in the populations subject to a strong hatchery influence.

To understand the potential significance of variation in SMR to growth, differentiation between species, and patterns in habitat use along productivity gradients that correspond to an energy-maximizing vs. cost-minimizing strategy, we tested whether there were predictable differences in metabolism and performance between juvenile steelhead trout and coho salmon. In addition to growth and SMR, we also measured and compared metrics of active physiology and swimming performance, including maximal metabolic rate (MMR) and aerobic scope (AS) for both wild and hatchery juvenile steelhead and coho under high- and low-food conditions (i.e. over a productivity gradient). Aerobic scope is the difference between SMR and MMR and represents the metabolic scope for activity, with high AS and MMR being correlates of increased athletic ability and swim performance (Plaut 2001; Killen *et al.* 2007). Our three main objectives were (i) to understand the differences in physiology underlying adaptive trade-offs between growth and other performance attributes along a productivity gradient, (ii) to determine whether differences in swimming performance matched differences in habitat use associated with energy-maximizing vs. cost-minimizing strategies, and (iii) to determine whether differences in physiology (SMR, MMR and AS), swim performance (UCrit) and growth between steelhead and coho are reduced under selection in a high productivity hatchery environment.

Our expectations were that steelhead and coho would exhibit a trade-off between growth and metabolism along a productivity gradient, with steelhead growing faster than coho at high-food abundance but more slowly at low food while exhibiting a higher SMR than coho and a higher UCrit associated with adaptation to a faster velocity habitat. Divergence between species in these complementary traits would provide strong evidence for adaptive trade-offs that match species to local habitat and energy availability.

Materials and methods

FISH COLLECTION AND REARING

Hatchery steelhead and coho salmon were obtained from Chapman Creek fish hatchery in Sechelt, BC, Canada (UTM 448100E 5478100N) on 12 August 2009. Average initial weights of hatchery steelhead and coho salmon were 1.25 ± 0.36 and 2.97 ± 0.73 g SD, respectively. Wild steelhead trout and coho salmon were captured using a combination of seining, minnow trapping and dip netting from McNab Creek, Port Mellon, BC, Canada (UTM 471738E 5490574N) on 28 August 2009. Fish collected from McNab Creek are presumed to be of wild origin because there is no history of hatchery releases into the creek. Average weights of wild steelhead and coho salmon were 0.74 ± 0.29 and 1.95 ± 0.60 g SD, respectively. Fish were transported to Chapman Creek hatchery and marked with visible implant elastomer (Northwest Marine Technology, Inc., Shaw Island, Washington, USA) so that SMR, MMR, AS, UCrit and growth data could be collected on individuals. Because hatchery fish were larger and available earlier than wild fish, the hatchery fish growth experiment was set up 16 days earlier than the wild fish growth experiment.

Fish were held indoor at Chapman Creek hatchery in two approximately 500 L flow-through aluminium troughs supplied with water from Chapman Creek. To optimize space, troughs were divided transversely into four serial compartments using 1-mm mesh partitions, which allowed hatchery and wild fish to be separated but reared in the same trough. Food, however, was too large to pass through the screen divider ensuring fish received the appropriate quantity of food. During the pre-treatment acclimation period, fish were fed *ad libitum* with Aqua Pride trout 48:16 pellets (Unifeed, Okotoks, AB, Canada; crude protein 48.0%, crude fat 16.0%, crude fibre 5.0%, sodium 0.40%, calcium 2.5% and phosphorous 1.5%). Average temperature for the hatchery fish growth experiment was $12.3 \text{ }^{\circ}\text{C} \pm 2.0$ SD (26th August–20th September) and $8.3 \text{ }^{\circ}\text{C} \pm 1.2$ SD for the wild growth experiment (21st September–11th October), which took place later because of the smaller size at age of wild fish. Hatchery fish were acclimated for approximately 2 weeks before food treatments were applied, whereas wild fish were acclimated for approximately 3 weeks to ensure complete transition to eating pelleted food. Fish were held under a 12L : 12D light cycle. All fish were weighed and measured to the nearest 0.01 g and 0.1 cm and were healthy and feeding prior to experimentation.

EXPERIMENTAL PROTOCOLS

Our experimental design included high- and low-food treatments for both wild and hatchery fish of both species ($n = 8$ experimental units in total). Twelve fish were initially stocked in each experimental unit (trough subsection) in both the hatchery and wild fish experiment. After the initial acclimation period, fish were measured to the nearest 0.01 g and 0.1 cm and placed on either high food (satiation for 3 weeks) or low food (1% body wt per day for 2 weeks followed by 7 days of starvation). We chose these extremes in ration to maximize contrast and the likelihood of detecting species differences. Seven days of starvation was used as the low-food treatment as this has been shown to be enough time to elicit a reduction in SMR (Van Leeuwen 2010). High-food fish were fed three times daily throughout the experiment and the low-food fish were fed once in the morning throughout the experiment prior to the 7-day starvation period. We estimated a satiation ration using the equation from Brett (1971) for juvenile sockeye salmon {consumption, % body dry weight = $14.5 - [5.15 \times \log_{10}(\text{wet weight of fish, g})]$ } to ensure that all fish received the same level of food based on the allometry of food consumption (smaller fish have a lower absolute ration than larger fish to achieve an equal degree of fullness, but satiate at a much higher per cent of body weight). We also used a bioenergetic model (Sullivan *et al.* 2001) to back calculate estimated food consumption based on growth at the observed rearing temperatures for both steelhead and coho as a form of *a posteriori* validation of ration. Based on these estimates, steelhead and coho at high food were growing at a rate consistent with satiation, thereby controlling for any potential differences in maximum food consumption between species.

Of the 12 fish stocked in each treatment at the start of the 3-week growth period, 7–11 were available for final analysis because a number of fish died, escaped during transfer to metabolic chambers, failed to grow, or showed obvious signs of poor health and were excluded from analysis. A subset of 6–8 steelhead trout and coho salmon from the remaining fish in each treatment were haphazardly selected for SMR, MMR and swim-performance trials. Time constraints prevented measuring metabolism and performance on all fish, but all successfully reared fish (7–11 per treatment) were used for growth-rate measurements ($n = 19$ for hatchery coho, 19 for hatchery steelhead and $n = 14$ for wild coho, 15 for wild steelhead). If individual fish

failed to complete any of the metabolism and performance measurements or were no longer identifiable (because of an elastomere mark that was no longer visible), they were also excluded from the analysis.

MEASURING STANDARD METABOLIC RATE (SMR)

Troughs were vacuum siphoned to remove food and debris the night before fish were placed in respirometry chambers. This ensured that fish were unfed for 20 h prior to oxygen uptake measurements and had sufficient time to evacuate their guts; 20 h post-feeding has been shown to be adequate for the specific dynamic action (SDA) response to subside (McCarthy 2000; Cutts, Metcalfe & Taylor 2002). SDA is an elevation in metabolic rate from the increased energy demands associated with digestion, immediately following a meal (Kleiber 1961; Jobling 1981; Alsop & Wood 1997), and is generally not considered part of SMR.

Respiration was measured using flow-through respirometry by placing individual fish into two separate glass respirometry chambers 9 h prior to oxygen consumption measurements, allowing measurements of SMR on two fish each day. Respirometers were constructed of 13.0 cm lengths of 2.8 cm diameter glass tubing, covered in black plastic to minimize fish activity during respirometry measurements (Cutts, Metcalfe & Taylor 2002). Glass respirometers were used to prevent possible issues with use of plastics (Stevens 1992). Measurements of background oxygen consumption for individual respirometry measurements were found to be negligible. After a minimum 9-h acclimation in the respirometer at constant flow, oxygen consumption was measured continuously overnight using a fibre optic oxygen meter (Foxy-Or125 oxygen sensors; Ocean Optics, Dunedin, Florida, USA) calibrated daily using nitrogen and aerated water at 13 °C. An air stone in the header tank of the respirometer apparatus kept inflow water fully saturated with oxygen. Flow to each respirometer was adjusted using micro valves to ensure that there was at least a 10% drop in oxygen tension between the inlet and outlet of each respirometer. Flow was measured by collecting the outlet water for a period of 60 s and weighing to the nearest 0.01 g periodically throughout the experiment. Temperature was held constant at 13.1 ± 0.4 °C SD in this and all other metabolic protocols, as continuously recorded using a submersible temperature logger.

The rate of oxygen consumption was determined using the following equation (Ege & Krough 1914):

$$\dot{M}O_2 = V_w \Delta C_w O_2$$

where V_w is the flow of water through the respirometer and $\Delta C_w O_2$ is the difference between the oxygen tension between water entering and leaving the respirometer. Concentration of oxygen was calculated by correcting PO_2 (partial pressure oxygen) for barometric pressure and multiplying by αO_2 ($\mu\text{mol L}^{-1} \text{ torr}^{-1}$), the solubility coefficient at the observed temperature. Continuous oxygen consumption measurements were averaged over half hour periods and plotted graphically. Periods of complete rest were readily discriminated from spontaneous activity, which appeared as distinct spikes in SMR. SMR was estimated using the lowest 1 h duration oxygen consumption values observed during the respirometry trial.

MEASURING MAXIMAL METABOLIC RATE (MMR)

After SMR was measured overnight, fish were transferred into two identical mini swim tunnels (Loligo Systems, Tjele, Denmark) and acclimated for 1 h at very low velocity (c. 0.3 BL s^{-1}). Black plastic was placed over the middle of the swim tunnel to allow an area of refuge where fish normally held during swimming trials. Fish were made

to swim at different velocities by adjusting the speed of a small motor driving an impeller built into one end of the swim tunnel. An air stone in the holding tank of the swim tunnel kept water fully saturated with oxygen. Water was periodically exchanged to prevent build up of nitrogenous waste. Measurements of background oxygen consumption during individual respirometry measurements were found to be negligible.

Swim tunnels were used to swim the fish to exhaustion, in order to determine MMR. Oxygen consumption of fish swum to exhaustion is believed to be a close approximation of MMR, which was measured using closed respirometry during swimming. Oxygen consumption was measured after 10 min of prolonged swimming at each velocity using a fibre optic oxygen meter (Foxy-Or125 oxygen sensors; Ocean Optics, Dunedin, Florida, USA). A reduction in oxygen tension of 20% was used as a threshold to terminate oxygen measurements and flush the tunnel with oxygenated water. Stepwise swimming at higher velocities was repeated until fish began to show signs of fatigue and needed to be repeatedly encouraged to swim using a bright light or by reversing the flow. At this point, oxygen consumption rate was measured until the fish could no longer swim, resulting in a decrease in oxygen consumption and termination of the swimming trial. The highest oxygen consumption rate usually corresponded to the highest water velocity in which the fish swam for a complete time interval and was used to estimate MMR. The rate of oxygen consumption was determined using the following equation: (Ege & Krough 1914)

$$\dot{M}O_2 = \frac{V_w \Delta C_w O_2}{\Delta t}$$

where V_w is the volume of water in the respirometer, $\Delta C_w O_2$ is the change in oxygen tension of the water and Δt , the time period associated with the drop in oxygen tension within the tunnel (Steffensen 1989). Concentration of oxygen was calculated as described earlier for SMR.

MEASURING SWIM PERFORMANCE (UCRIT)

After MMR trials were completed, fish were subjected to a UCrit swim performance test, which is a common way to assess differences in endurance swimming and has been used in a number of studies (Taylor & McPhail 1985; Lee *et al.* 2003; Fanguie *et al.* 2008) as outlined in Brett (1964). We chose to assess aspects of endurance swimming as opposed to burst swimming as it was hypothesized that endurance swimming would be important for fish adapted to holding in faster velocity habitats. Immediately following the MMR trial, fish were placed into a larger swim tunnel (Loligo Systems) and acclimated at a velocity of 0.3 BL s^{-1} for a minimum of 3 h prior to the UCrit test. This provided adequate time for recovery (Kolok 1991; Jain, Birtwell & Farrell 1998) from the previous MMR trial that was used as a pre-swim. Pre-swims are commonly used to familiarize fish with swimming and have been used in a number of studies (Taylor & McPhail 1985; Lee *et al.* 2003; Fanguie *et al.* 2008). In order to allow all metabolic and performance measurements to be carried out in a single day, fish were swum in pairs (one steelhead and one coho) in a Beamish-style swim tunnel (Loligo Systems). Although fish swam in pairs, agonistic behaviours were not observed, nor were there differences in swim performance between fish swum separately or together (determined prior to experimentation on several test fish).

Velocity of water inside the tunnel was controlled by a rheostat and calibrated using a flow meter. During acclimation and the swim performance test, $\frac{3}{4}$ of the swim tunnel was covered in black plastic to provide cover and lower stress of test fish. For the UCrit test, water velocity was increased by 0.3 BL s^{-1} every 10 min in a stepwise fashion,

as outlined by Fague *et al.* (2008), until fish fatigued. Fish were considered fatigued and the test terminated if three repeated attempts to force the fish to continue swimming were not successful. UCrit was calculated using the following equation from Brett (1964), which calculates the final endurance swimming speed at fatigue by weighting the speed of the two final swim steps by their duration (in min):

$$\text{UCrit} = U_i + [U_{ii}(t_i/t_{ii})]$$

where U_i is the highest speed the fish swam during the full time period, U_{ii} is the incremental speed increase, t_i is the time (fraction of the final time step) at which the fish swam at the final speed and t_{ii} is the prescribed period of swimming at each velocity (10 min). Because of the small size of the fish in relation to the cross-sectional areas of the swim tunnel, UCrit values were not corrected for a solid blocking effect.

STATISTICAL ANALYSES AND CALCULATIONS

Instantaneous growth rates of fish (% per day) were calculated as $\{[\log_e(\text{final mass}) - \log_e(\text{initial mass})]/\text{time (days)}\} \times 100$ (Ricker 1975). AS was determined by subtracting MMR from SMR.

All data, except UCrit, were log transformed to linearize the data and meet assumptions of normality and homogeneity of variance. Growth data was log transformed using a constant (1.60 for wild fish and 1.75 for hatchery fish) to allow transformation of negative growth values associated with the low-food treatments.

We tested for the effects of food, individual fish mass, and species on SMR, MMR, AS, UCrit and growth using analysis of covariance (ANCOVA), including interactions between food and mass, food and species, and mass by species. Interaction terms and independent variables that were not significant at $P < 0.05$ were removed from each model. Absolute, rather than mass-specific, values for SMR, MMR, AS, UCrit and growth were used with mass as a covariate because of the large variation in fish size. Our analysis implicitly assumes that all log-log relationships are linear. This is an important consideration because of the limited overlap in size between steelhead and coho, and is reasonably well supported by linear log-log relationships in earlier studies of juvenile salmonid metabolism (for both SMR and MMR) over size ranges that bracket those in this study (e.g. Wieser 1985; Cutts, Metcalfe & Taylor 2002).

Data for all food treatments and species were first analysed together. If a significant food by species interaction was found, steelhead and coho data were analysed separately using ANCOVA with food, individual fish mass, and the interaction between food and individual fish mass as factors. All analysis was conducted using R version 2.8.1 statistical software (R Development Core Team, Vienna, Austria).

Results

Mean mass of wild steelhead and coho ($\pm$ SD) at the end of the experiment was 1.70 ± 0.32 and 3.89 ± 0.56 g at high food and 0.92 ± 0.27 and 2.92 ± 2.23 g for the low-food treatment, respectively. Mean mass of hatchery steelhead and coho ($\pm$ SD) at the end of the experiment was 3.33 ± 0.60 and 5.40 ± 1.28 g for high-food treatment and 1.64 ± 0.60 and 3.50 ± 0.87 g for the low-food treatment, respectively. Mass was highly significant in all models (Table 1).

STANDARD METABOLIC RATE (SMR)

The high-food treatment substantially elevated SMR for both wild steelhead and wild coho (Fig. 1, Table 1), but high

food significantly elevated SMR only for steelhead ($F_{2,10} = 5.68$, $P = 0.04$) among the hatchery fish (Fig. 1, Table 1).

Contrary to expectation, there was no difference in SMR between wild steelhead and coho ($F_{4,22} = 2.92$, $P = 0.10$). However, there was a significant interaction between food and species for hatchery fish ($F_{4,22} = 17.8$, $P < 0.001$), with hatchery steelhead having a significantly higher SMR than hatchery coho ($F_{2,10} = 5.68$, $P = 0.04$) at high food, but a significantly lower SMR than hatchery coho at low food ($F_{2,11} = 5.55$, $P = 0.04$).

MAXIMAL METABOLIC RATE (MMR)

Steelhead had a significantly higher MMR than coho for both wild ($F_{5,25} = 53.6$, $P < 0.001$) and hatchery fish; ($F_{4,23} = 5.56$, $P = 0.03$; Fig. 2, Table 1).

AEROBIC SCOPE (AS)

Wild steelhead had a significantly higher AS than wild coho at both high and low food ($F_{2,12} = 10.09$, $P = 0.008$). The high-food treatment significantly decreased AS in wild fish ($F_{2,12} = 9.84$, $P = 0.004$; Table 1, Fig. 3), but there was no significant effect of food or species on AS in hatchery fish (Table 1, Fig. 3).

SWIM PERFORMANCE (UCRIT)

Wild steelhead had a significantly higher UCrit than wild coho ($F_{4,25} = 18.1$, $P < 0.001$; Table 1, Fig. 4). There was no significant food effect on UCrit (Table 1; Fig. 4) or difference between species for hatchery steelhead and coho ($F_{4,24} = 0.03$, $P = 0.87$).

INSTANTANEOUS GROWTH RATE

Mean growth rates of wild steelhead and coho were $3.08 \pm 0.33\%$ and $1.91 \pm 0.68\%$ body wt per day at high-food and $-1.01 \pm 0.40\%$ and $-1.00 \pm 0.25\%$ body wt per day for the low-food treatments, respectively. Mean growth rates of hatchery steelhead and coho ($\pm$ SD) were $3.35 \pm 0.50\%$ and $1.79 \pm 0.45\%$ body wt per day at high-food and $-0.90 \pm 0.29\%$ and $-0.96 \pm 0.37\%$ body wt per day for the low-food treatments, respectively. Instantaneous growth rate of steelhead was significantly higher than coho growth in both the wild ($F_{3,25} = 9.22$, $P = 0.006$) and hatchery fish ($F_{3,37} = 22.1$, $P < 0.001$; Table 1, Fig. 5). As expected, fish on the high-food ration also grew faster.

Discussion

Contrary to expectation, SMR of wild steelhead was not higher than wild coho at either food level, providing little evidence for a trade-off between SMR and growth along a productivity gradient. Our results also demonstrate that the quantity of food consumed directly affects the magnitude of

Table 1. The effect of food level, species, mass (g) and mass $\times$ species, and species $\times$ food interactions on physiological parameters: SMR ($\mu\text{mol h}^{-1}$), MMR ($\mu\text{mol h}^{-1}$), AS ($\mu\text{mol h}^{-1}$), UCrit (cm s^{-1}) and growth rate (% body wt per day) for both wild (W) and hatchery (H) steelhead (St) and coho (Co) at high (H) and low (L) food

Dependent variable	Fish type	Log mass (x)	Length	Food level	Species	Log mass $\times$ food	Log mass $\times$ species	Species $\times$ food	Intercept	n	F	P value
Log SMR	H	$0.52 \times \text{Log mass}$		-0.05 (L) 0 (H)	0.10 (St) 0 (Co)	-	-	-0.22 (StL) 0 (StH), (CoL), (CoH)	0.95	27	59.56	<0.001
	W	$0.77 \times \text{Log mass}$		-0.26 (L) 0 (H)	-	-	$-0.41 \times \text{Log mass (StH), (StL)}$ 0 (CoH), (CoL)	-	0.85	27	90.11	<0.001
Log MMR	H	$0.95 \times \text{Log mass}$		-	0.09 (St) 0 (Co)	-	-	-	1.15	27	105.70	<0.001
	W	$1.64 \times \text{Log mass}$		0.39 (L) 0 (H)	0.54 (St) 0 (Co)	$-0.59 \times \text{Log mass (StL, CoL)}$ 0 (StH), (CoH)	$-0.50 \times \text{Log mass (StH), (StL)}$ 0 (CoH), (CoL)	-0.36 (StL) 0 (StH), (CoL), (CoH)	0.70	27	42.08	<0.001
Log AS	H	$0.90 \times \text{Log mass}$		-	-	-	-	-	1.05	27	60.52	<0.001
	W	$1.17 \times \text{Log mass}$		0.16 (L) 0 (H)	-	-	-	-	0.73	27	18.81	<0.001
UCrit	H		$3.53 \times \text{Length}$	-	-	-	-	-	17.55	27	23.37	<0.001
	W		$3.08 \times \text{Length}$	-	5.04 (St) 0 (Co)	-	-	-	16.45	27	4.80	0.02
Log growth	H	$0.53 \times \text{Log mass}$		-0.55 (L) 0 (H)	0.21 (St) 0 (Co)	-	-	-	0.20	38	186.10	<0.001
	W	$0.74 \times \text{Log mass}$		-0.73 (L) 0 (H)	0.34 (St) 0 (Co)	-	-	-	-	29	60.86	<0.001

SMR, standard metabolic rate; MMR, maximal metabolic rate; AS, aerobic scope.

StH, StL, CoH and CoL represent steelhead high food, steelhead low food, coho high food and coho low food treatments, respectively. Parameters for significant variables are presented. Sample equation for predicting SMR for hatchery coho salmon under high food is as follows: $\text{SMR} = 0.52 \times (\text{log mass}) + 0.95$.

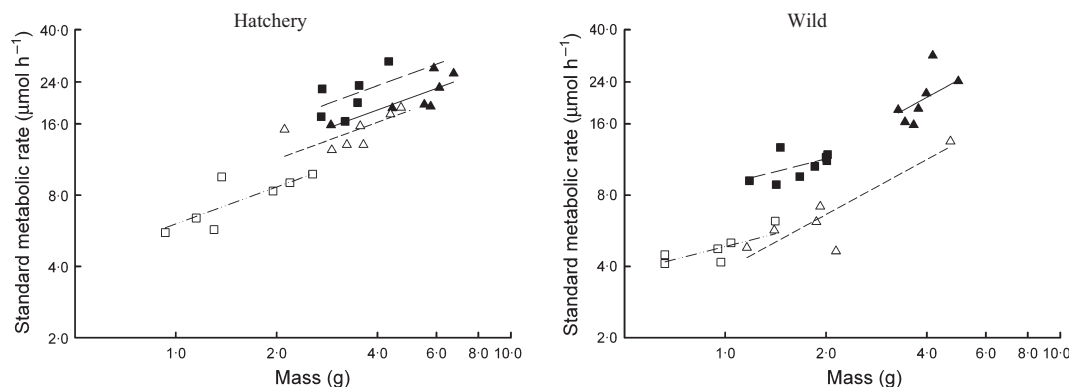


Fig. 1. The relationship between standard metabolic rate and mass for juvenile steelhead (squares) and coho salmon (triangles) under high-food (closed) and low-food (open) treatments. Figure on left represents the hatchery population, and figure on right represents wild fish. See text for statistical analysis and Table 1 for parameter significance.

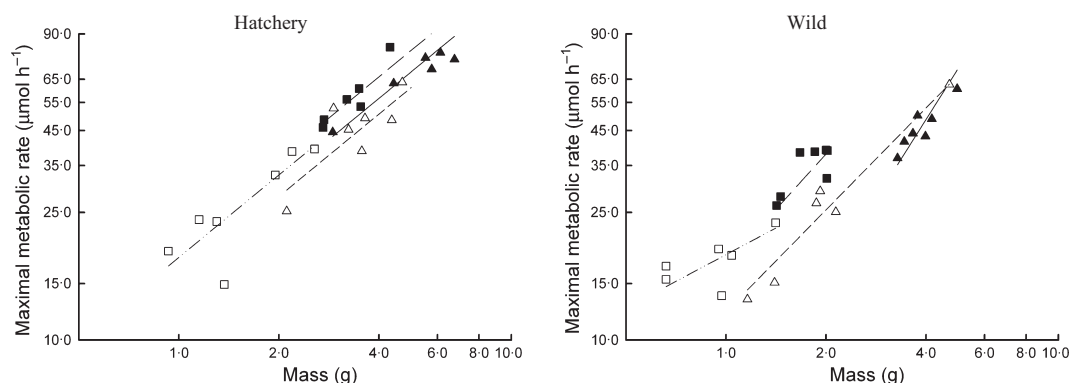


Fig. 2. The relationship between maximal metabolic rate and mass for juvenile steelhead (squares) and coho salmon (triangles) under high-food (closed) and low-food (open) treatments. Figure on left represents the hatchery population, and figure on right represents wild fish. See text for statistical analysis and Table 1 for parameter significance.

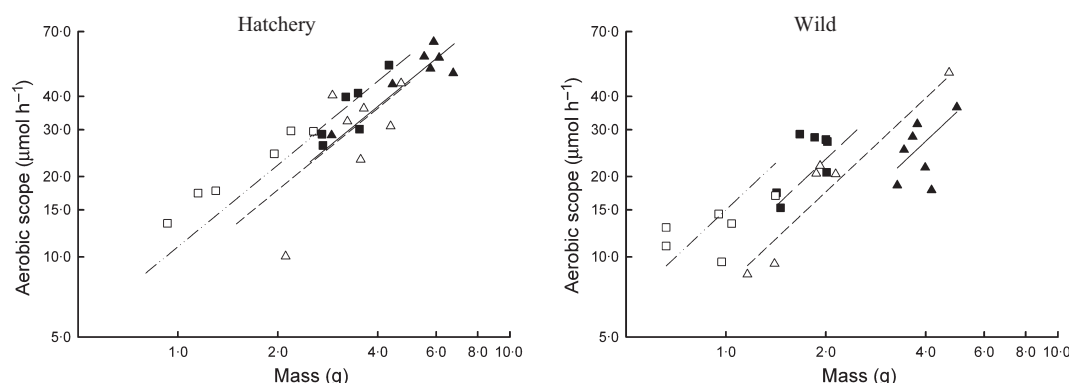


Fig. 3. The relationship between aerobic scope and mass for juvenile steelhead (squares) and coho salmon (triangles) under high-food (closed) and low-food (open) treatments. Figure on left represents the hatchery population, and figure on right represents wild fish. See text for statistical analysis and Table 1 for parameter significance.

SMR similarly for both steelhead and coho, with higher food consumption and growth leading to higher SMR.

Earlier studies have shown an adaptive trade-off involving growth and swimming performance (U_{crit}) between populations of Atlantic silverside (*Menidia menidia*) along a latitudinal gradient (countergradient variation; Conover & Present

1990), with faster growing northern populations exhibiting lower swimming capacity and higher vulnerability to predation (Billerbeck, Lankford & Conover 2001; Lankford, Billerbeck & Conover 2001; Chiba, Arnott & Conover 2007). In contrast, southern populations, with a longer growing season, have lower growth rates and an increased swim perfor-

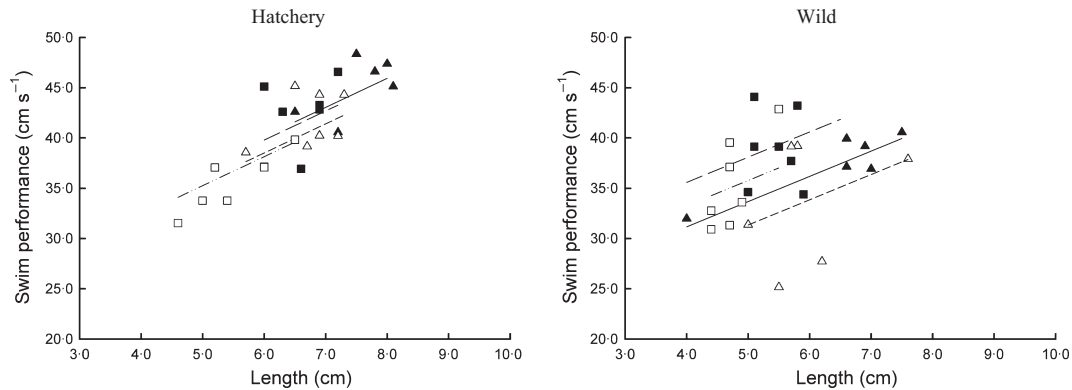


Fig. 4. The relationship between swim performance and length for juvenile steelhead (squares) and coho salmon (triangles) under high-food (closed) and low-food (open) treatments. Figure on left represents the hatchery population, and figure on right represents wild fish. See text for statistical analysis and Table 1 for parameter significance.

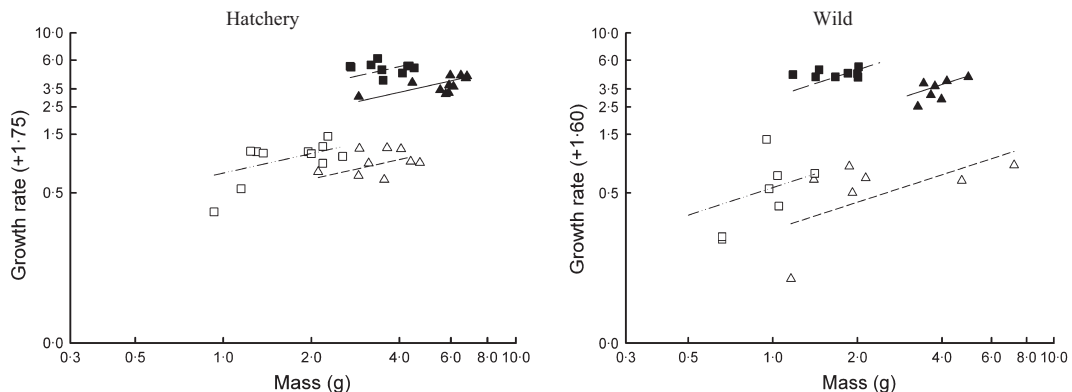


Fig. 5. The relationship between growth rate and mass for juvenile steelhead (squares) and coho salmon (triangles) under high-food (closed) and low-food (open) treatments. Figure on left represents the hatchery population, and figure on right represents wild fish. See text for statistical analysis and Table 1 for parameter significance.

mance associated with lower mortality rates. This trade-off between higher growth and lower swimming capacity was not evident between wild steelhead and wild coho in our study; wild steelhead grew faster than wild coho and also appeared to have a higher UCrit, AS and MMR.

Maximal metabolic rate is the maximum amount of energy that a fish can respire aerobically and is primarily limited by the ability of the cardiovascular system to deliver oxygen to mitochondria in the muscles. Wild steelhead had a higher MMR than wild coho, resulting in a higher AS as SMR was similar between species. Conceivably, a higher AS may provide greater excess available metabolic capacity for defending territories, swimming, and active prey capture leading to greater food consumption and growth. Our results suggest that reduced swim performance, associated with higher growth in northern populations showing countergradient variation, may be the result of a lower AS caused by elevated SMR at high growth and food consumption levels, or a greater investment in a larger digestive system at the cost of reduced investment in muscle or cardiovascular tissue that enhances swimming performance (Alvarez & Metcalfe 2007).

A low growth efficiency, as manifest by juvenile wild steelhead (e.g. Sullivan *et al.* 2001), may appear non-adaptive on

first consideration. However, reduced growth efficiency may become adaptive at high food availability (Sibly 1981), as a high food consumption/short gut residence strategy may maximize energy assimilation per unit time if food quality is high (Millidine, Armstrong & Metcalfe 2009). Digestion has been shown to be the bottleneck that minimizes energy assimilation when food is in excess (Booth 1990; Hart & Gill 1992), thereby limiting growth. Conceivably, an adaptive trade-off between growth efficiency and growth rate could exist along a productivity gradient (e.g. Finstad *et al.* 2011). A fixed, low growth efficiency based on high food consumption and rapid extraction of labile energy would become maladaptive in resource-poor environments, where slower more efficient assimilation would be favoured. Such contrasts in the efficiency of energy metabolism between species have parallels in unicellular organisms, which switch to lower efficiency metabolic pathways to maximize growth at high resource availability (Molenaar *et al.* 2009). Slower coho growth could also reflect greater energy costs associated with a higher frequency of aggressive interactions relative to steelhead (Puckett & Dill 1985; Young 2001), but this effect appears insufficient to reduce coho growth efficiency below that of steelhead (Sullivan *et al.* 2001).

Differences in physiology between wild steelhead and wild coho appear consistent with energy-maximizing vs. cost-minimizing strategies (Arendt 1997; Arendt & Wilson 1997; Tucker & Rasmussen 1999). A higher MMR, AS and swim performance make wild steelhead better suited for holding in faster velocity habitats compared with wild coho, although both species grow faster in pool habitats where swimming costs are low, provided that prey are abundant (Young 2001). When preferred pool habitats are occupied by larger dominant coho, holding in faster velocity habitats may allow steelhead, with a lower growth efficiency and potentially shorter gut residence time, the ability to maximize food consumption through higher prey encounter rates in riffles relative to pools; similar use of higher velocity habitats to maximize prey encounter rates have been noted for other salmonids [e.g. Atlantic salmon (Nislow, Folt & Parrish 1999) and brown trout (Hayes, Stark & Shearer 2000)]. This strategy may allow wild steelhead to offset the cost of a shorter growing season because of a later emergence time compared with wild coho (steelhead are spring spawners, coho are fall spawners whose fry hatch 4–6 weeks earlier than steelhead and establish territories in preferred pool habitat). Like northern temperate populations of Atlantic silversides, salmonids are also under strong selection to exceed a threshold body size for overwintering (Post & Evans 1989; Shuter & Post 1990; Schultz & Conover 1997; Post & Parkinson 2001); however, wild steelhead with a higher MMR may be able to compensate for an elevated SMR associated with increased food consumption and high growth rates while still maintaining swim performance, unlike northern populations of Atlantic silverside.

Hatcheries represent a resource-rich environment and generally cause selection for higher growth rates (Petersson *et al.* 1996; Fleming *et al.* 2002; Sundstrom, Lohmus & Devlin 2005) and therefore indirectly higher SMR, as SMR is correlated with increased growth rate and food consumption (Van Leeuwen 2010). Our observation of reduced differences in SMR, AS, growth and swim performance between hatchery steelhead and coho is consistent with convergent selection in a relatively homogenous hatchery environment where both species are raised at low velocities and high productivity over multiple generations.

Elevated SMR leading to decreased AS in hatchery steelhead may underlie reduced differences in swimming performance between hatchery steelhead and coho. This is consistent with other studies (e.g. Brett, Hollands & Alerdice 1958; Bams 1967) which have shown that hatcheries select for higher growth rate at the cost of reduced swimming performance. It is also consistent with the countergradient variation trade-offs described earlier and studies that have documented a trade-off between SMR and swimming performance in brown trout populations along a thermal gradient (Cano & Nicieza 2006) and suggests that a trade-off between growth rate and swimming ability may be intrinsic to growth variation among populations within many species. The only caveats for concluding that differences between species were more pronounced in wild fish is that the wild fish experiment

occurred 3 weeks later at a lower temperature (12.3 vs. 8.3 °C) than the experiment with hatchery origin fish. Because steelhead are somewhat more cold adapted, a lower temperature could enhance differences in performance between the species, although all metabolic tests were performed at a fixed temperature of 13.0 °C where temperature-related differences in performance between species should have been minimal. In addition, lack of replication of hatchery and wild populations means that inter-population variation in metabolic optima may contribute to the inferred hatchery influence. However, the fact that differences between streams are consistent with hatchery effects makes this less likely.

In contrast, interspecific differences between coho and steelhead that allow higher maximum steelhead growth do not appear to involve a trade-off between growth rate and swimming ability, but rather between growth rate and growth efficiency. Bioenergetic models (e.g. Sullivan *et al.* 2001) and laboratory rearing experiments under low activity (Wurtsbaugh & Davis 1977) predict that steelhead have a much lower growth efficiency than coho, and have to consume approximately 50% more food to achieve equivalent growth. This suggests that adaptive trade-offs between these species involve different digestion strategies that influence growth efficiency, where steelhead may have a high food consumption strategy that strips out more labile energy with lower efficiency than coho, allowing them to achieve higher growth at high food abundance. Different digestive strategies to maximize energy assimilation are common among vertebrates (Milton 1981; Hume 1989; van Gils *et al.* 2007), but remain poorly documented among fish (see Nicieza, Reiriz & Brana (1994) and Millidine, Armstrong & Metcalfe (2009) for exceptions), but we suggest that they may underlie many of the adaptive trade-offs related to growth that allow for differentiation and coexistence of ecologically similar species like salmonids.

This apparent adaptive trade-off between high growth and low growth efficiency has also been demonstrated in other species and ecotypes of salmonid. Similar to the contrast in habitat use and growth efficiency observed between coho and steelhead, juvenile anadromous brook trout have a more streamlined body shape, occupy higher velocity habitats and show increased food consumption and lower growth efficiencies than sympatric resident juveniles (Morinville & Rasmussen 2003, 2008). Forseth *et al.* (1999) demonstrated a similar pattern between resident and migratory juvenile Atlantic salmon. Tucker & Rasmussen (1999) also showed that Atlantic salmon have higher food consumption, growth and lower growth efficiency than sympatric brook trout (although they attributed lower growth efficiency to greater costs of foraging in high-velocity habitats). This suggests that adaptive trade-offs between species (steelhead vs. coho, Atlantic salmon vs. brook trout) and ecotypes (anadromous vs. resident) of sympatric salmonids are similar (e.g. energy maximizers vs. cost minimizers), but differ from those between populations on a latitudinal gradient (growth vs. swimming ability trade-off).

One limitation of this study is the difference in size between species and between hatchery and wild fish. Difference in emergence times between steelhead and coho make it difficult to size match fish without reducing coho growth by rearing at lower temperatures or under food deprivation until steelhead are of similar size (Young 2001), which could introduce potential artefacts (e.g. compensatory growth), as could selecting steelhead or coho from different ends of their size distributions.

While conclusions with respect to metabolic rates are size-corrected and therefore robust, our growth results should be interpreted with caution because steelhead were smaller than coho and the allometry of growth predicts higher growth rates of smaller fish; consequently, slopes of the lines in Fig. 5 should be negative. The positive slopes in Fig. 5 indicate the presence of dominance hierarchies where larger fish monopolize resources, even in high-food treatments. Higher steelhead growth in our study is confounded with smaller body size, making it difficult to unambiguously attribute higher steelhead growth to a species effect, although higher growth rates of steelhead relative to coho are well documented elsewhere (Hartmann 1965; Sullivan *et al.* 2001; Young 2001).

This study does not support a simple trade-off between SMR and growth between juvenile steelhead and coho metabolism, although juvenile steelhead and coho do appear to adopt contrasting energy maximizing and minimizing strategies. The most distinctive emergent trade-off between wild steelhead and coho appears to involve growth efficiency, similar to the metabolic trade-off which appears to exist between brook trout ecotypes adopting anadromous and resident life histories (Morinville & Rasmussen 2003, 2008), Atlantic salmon vs. brook trout (Tucker & Rasmussen 1999), and brown trout vs. Arctic char (Finstad *et al.* 2011), suggesting that this strategy may be fairly general among salmonids. The mechanisms underlying variation in growth efficiency are unclear and have been previously attributed to differences in costs of activity (Tucker & Rasmussen 1999) or aggression (Finstad *et al.* 2011). Divergent digestive strategies, which are well documented among terrestrial vertebrates (Milton 1981; Hume 1989; van Gils *et al.* 2007), however, represent an additional mechanism and may be an important but overlooked aspect of adaptive strategies of juvenile salmonids and fish in general.

Acknowledgements

We thank two anonymous reviewers for helpful comments that greatly improved the manuscript. We also thank Dr Dave Bates for advice and assistance with stream selection and fish collection as well as Bob Anstead, members of the Sunshine Coast Salmon Enhancement Society and Chapman Creek hatchery for providing a place to carry out research. Funding was provided by an NSERC discovery grant to JSR.

References

Alsop, D.H. & Wood, C.M. (1997) The interactive effects of feeding and exercise on oxygen consumption, swimming performance and protein usage in juvenile rainbow trout (*Oncorhynchus mykiss*). *The Journal of Experimental Biology*, **200**, 2337–2346.

- Alvarez, D., Cano, J.M. & Nicieza, A.G. (2006) Microgeographic variation in metabolic rate and energy storage of brown trout: countergradient selection or thermal sensitivity? *Evolutionary Ecology*, **20**, 345–363.
- Alvarez, D. & Metcalfe, N.B. (2007) The tradeoff between catch-up growth and escape speed: variation between habitats in the cost of compensation. *Oikos*, **116**, 1144–1151.
- Alvarez, D. & Nicieza, A.G. (2005) Is metabolic rate a reliable predictor of growth and survival of brown trout (*Salmo trutta*) in the wild? *Canadian Journal of Fisheries and Aquatic Sciences*, **62**, 643–649.
- Arendt, J.D. (1997) Adaptive intrinsic growth rates: an integration across taxa. *Quarterly Review of Biology*, **72**, 149–177.
- Arendt, J.D. & Wilson, D.S. (1997) Optimistic growth: competition and an ontogenetic niche-shift select for rapid growth in pumpkinseed sunfish (*Lepomis gibbosus*). *Evolution*, **51**, 1946–1954.
- Bams, R.A. (1967) Differences in performance of naturally and artificially propagated sockeye salmon migrant fry, as measured with swimming and predation tests. *Journal of the Fisheries Research Board of Canada*, **24**, 1117–1153.
- Billerbeck Jr, J.M., Lankford Jr, T.E. & Conover, D.O. (2001) Evolution of intrinsic growth and energy acquisition rates. I. Trade-offs with swimming performance in *Menidia menidia*. *Evolution*, **55**, 1863–1872.
- Biro, P.A. & Stamps, J.A. (2008) Are animal personality traits linked to life-history productivity? *Trends in Ecology & Evolution*, **23**, 361–368.
- Biro, P.A. & Stamps, J.A. (2010) Do consistent individual differences in metabolic rate promote consistent individual differences in behavior? *Trends in Ecology & Evolution*, **25**, 653–659.
- Bisson, P.A., Sullivan, K. & Nielsen, J.L. (1988) Channel hydraulics, habitat use, and body form of juvenile coho salmon, steelhead and cutthroat trout in streams. *Transactions of the American Fisheries Society*, **117**, 262–273.
- Bochdansky, A.B., Gronkjaer, P., Herra, T.P. & Leggett, W.C. (2005) Experimental evidence for selection fish larvae with high metabolic rates in a food limited environment. *Marine Biology*, **147**, 1413–1417.
- Boily, P. (2002) Individual variation in metabolic traits of wild nine-banded armadillos (*Dasypus novemcinctus*), and the aerobic capacity model for the evolution of endothermy. *Journal of Experimental Biology*, **205**, 3207–3214.
- Booth, D.J. (1990) Effect of water temperature on stomach evacuation rates and estimates of daily food intake of bluegill sunfish (*Lepomis macrochirus Rafinesque*). *Canadian Journal of Zoology*, **68**, 591–595.
- Bozinovic, S., Rojas, J.M., Broitman, B.R. & Vasquez, R.A. (2009) Basal metabolism is correlated with habitat productivity among populations of degus (*Octodon degus*). *Comparative Biochemistry and Physiology Part A*, **152**, 560–564.
- Brett, J.R. (1964) The respiratory metabolism and swimming performance of young sockeye salmon. *Journal of the Fisheries Research Board of Canada*, **21**, 1183–1226.
- Brett, J.R. (1971) Satiation time, appetite and maximum food intake of Sockeye salmon (*Oncorhynchus nerka*). *Journal of the Fisheries Research Board of Canada*, **28**, 409–415.
- Brett, J.R., Hollands, H. & Alderice, D.F. (1958) The effect of temperature on the cruising speed of young sockeye and coho salmon. *Journal of the Fisheries Research Board of Canada*, **15**, 587–605.
- Bugert, R.M. & Bjornn, T.C. (1991) Habitat use by steelhead and coho salmon and their responses to predators and cover in laboratory streams. *Transaction of the American Fisheries Society*, **120**, 486–493.
- Cano, J.M. & Nicieza, A.G. (2006) Temperature, metabolic rate, and constraints on locomotor performance in ectotherm vertebrates. *Functional Ecology*, **20**, 464–470.
- Careau, V., Reale, D., Humphries, M.M. & Thomas, D. (2010) The pace of life under artificial selection: personality, energy expenditure, and longevity are correlated in domestic dogs. *American Naturalist*, **175**, 753–758.
- Chiba, S., Arnott, S.A. & Conover, D.O. (2007) Coevolution of foraging behavior with intrinsic growth rate: risk taking in naturally and artificially selected growth genotypes of *Menidia menidia*. *Oecologia*, **15**, 237–246.
- Conover, D.O. & Present, T.C. (1990) Countergradient variation in growth rate: compensation for length of the growing season among Atlantic silver-sides from different latitudes. *Oecologia (Berlin)*, **83**, 316–324.
- Cutts, C.J., Metcalfe, N.B. & Taylor, A.C. (2002) Juvenile Atlantic Salmon (*Salmo salar*) with relatively high standard metabolic rates have small metabolic scopes. *Functional Ecology*, **16**, 73–78.
- Djawadan, M., Rose, M.R. & Bradley, T.J. (1997) Does selection for stress resistance lower metabolic rates? *Ecology*, **78**, 828–837.

- Ege, R. & Krough, A. (1914) On the relation between the temperature and the respiratory exchange in fishes. *International Review of Hydrobiology Hydrograph*, **7**, 48–55.
- Fangue, N.A., Mandic, M., Richards, J.G. & Schulte, P.M. (2008) Swimming performance and energetics as a function of temperature in Killifish *Fundulus heteroclitus*. *Physiological and Biochemical Zoology*, **81**, 389–401.
- Finstad, A.G., Forseth, T., Ugedal, O. & Naesje, T.F. (2007) Metabolic rate, behaviour and winter performance in juvenile Atlantic salmon. *Functional Ecology*, **21**, 905–912.
- Finstad, A.G., Forseth, T., Jonsson, B., Bellier, E., Hesthagen, T., Jensen, A.J., Hessen, D.O. & Foldvik, A. (2011) Competition exclusion along climate gradients: energy efficiency influences the distribution of two salmonid fishes. *Global Change Biology*, **17**, 1703–1711.
- Fleming, I.A., Agustsson, T., Finstad, B., Johnsson, J.I. & Björnsson, B.T. (2002) Effects of domestication on growth physiology and endocrinology of Atlantic salmon (*Salmo salar*). *Canadian Journal of Fisheries and Aquatic Sciences*, **59**, 1323–1330.
- Forseth, T., Naesje, T.F., Jonsson, B. & Harsaker, K. (1999) Juvenile migration in brown trout: a consequence of energetic state. *Journal of Animal Ecology*, **68**, 783–793.
- Fraser, F.J. (1969) Population density effects on survival and growth of juvenile coho salmon and steelhead trout in experimental stream channels. *Symposium on Salmon and Trout in Stream* (ed. T.G. Northcote), pp. 253–266. University of British Columbia Press, Vancouver, BC.
- Garland Jr, T. & Adolph, S.C. (1991) Physiological differentiation of vertebrate populations. *Annual Review of Ecology and Systematics*, **22**, 193–228.
- van Gils, J.A., Dekinga, A., van den Hout, P.J., Spaans, B. & Piersma, T. (2007) Digestive organ size and behavior of red knots (*Calidris canutus*) indicate the quality of their benthic food stocks. *Israel Journal of Ecology and Evolution*, **53**, 329–346.
- Gross, M.R. (1998) One species with two biologies: Atlantic salmon (*Salmo salar*) in the wild and in aquaculture. *Canadian Journal of Fisheries and Aquatic Sciences*, **55**(Suppl. 1), 131–144.
- Hart, P.J.B. & Gill, A.B. (1992) Constraints on prey size selection by the three-spined stickleback: energy requirements and the capacity of fullness of the gut. *Journal of Fish Biology*, **40**, 205–218.
- Hartmann, G.F. (1965) The role of behavior in the ecology and interaction of underyearling coho salmon (*Oncorhynchus kisutch*) and steelhead trout (*Salmo gairdneri*). *Journal of Fisheries Research Board of Canada*, **22**, 1035–1081.
- Hayes, J.W., Stark, J.D. & Shearer, K.A. (2000) Development and test of a whole-lifetime foraging and bioenergetics growth model for drift-feeding brown trout. *Transactions of the American Fisheries Society*, **129**, 315–332.
- Hume, I.D. (1989) Optimal digestive strategies in mammalian herbivores. *Physiological Zoology*, **62**, 1145–1163.
- Jain, K.E., Birtwell, I.K. & Farrell, A.P. (1998) Repeat swimming performance of mature sockeye salmon following a brief recovery period: a proposed measure of fish health and water quality. *Canadian Journal of Zoology*, **76**, 1488–1498.
- Jobling, M. (1981) The influences of feeding on the metabolic rate of fishes: a short review. *Journal of Fish Biology*, **18**, 385–400.
- Johnston, J.M. (1970) *Food and Feeding Habits of juvenile Coho Salmon and Steelhead Trout in Worthy Creek*. MSc Thesis. University of Washington, Seattle, WA.
- Killen, S.S., Costa, I., Brown, J.A. & Gamperl, K.A. (2007) Little left in the tank: metabolic scaling in marine teleosts and its implications for aerobic scope. *Proceedings of the Royal Society B*, **274**, 431–438.
- Kleiber, M. (1961) *The Fire of Life: An Introduction to Animal Energetics*. John Wiley and Sons, New York, London.
- Kolok, A.S. (1991) Photoperiod alters the critical swimming speed of juvenile largemouth bass, *Micropterus salmoides*, acclimated to cold water. *Copeia*, **1991**, 1085–1090.
- Lankford, T.E., Billerbeck Jr, J.M. & Conover, D.O. (2001) Evolution of intrinsic growth and energy acquisition rates. II. Trade-offs with vulnerability to predation in *Menidia menidia*. *Evolution*, **55**, 1873–1881.
- Larivee, M.L., Boutin, S., Speakman, J.R., McAdam, A.G. & Humphries, M.M. (2010) Associations between over-winter survival and resting metabolic rate in juvenile North American red squirrels. *Functional Ecology*, **24**, 597–607.
- Lee, C.G., Farrell, A.P., Lotto, A., MacNutt, M.J., Hinch, S.G. & Healey, M.C. (2003) The effect of temperature on swimming performance and oxygen consumption in adult sockeye (*Oncorhynchus nerka*) and coho (*O. kisutch*) salmon stocks. *The Journal of Experimental Biology*, **206**, 3239–3251.
- Lovegrove, B.G. (2003) The influence of climate on the basal metabolic rate of small mammals: a slow-fast metabolic continuum. *Journal of Comparative Physiology B*, **173**, 87–112.
- Mathot, K.J., Godde, S., Careau, V., Thomas, D.W. & Giraldeau, L. (2009) Testing dynamic variance-sensitive foraging using individual differences in basal metabolic rates of zebra finches. *Oikos*, **118**, 545–552.
- McCarthy, I.D. (2000) Temporal repeatability of relative standard metabolic rate in juvenile Atlantic salmon and its relation to life history variation. *Journal of Fish Biology*, **57**, 224–238.
- Metcalfe, N.B., Taylor, A.C. & Thorpe, J.E. (1995) Metabolic rate, social status and life-history strategies in Atlantic salmon. *Animal Behaviour*, **49**, 431–436.
- Millidine, K.J., Armstrong, J.D. & Metcalfe, N.B. (2009) Juvenile salmon with high standard metabolic rates have higher energy cost but can process meals faster. *Proceedings of the Royal Society B*, **276**, 2103–2108.
- Milton, K. (1981) Food choice and digestive strategies of two sympatric primate species. *The American Naturalist*, **117**, 496–505.
- Molenaar, D., van Berlo, R., de Ridder, D. & Teusink, B. (2009) Shifts in growth strategies reflect tradeoffs in cellular economics. *Molecular Systems Biology*, **5**, 323.
- Morinville, G.R. & Rasmussen, J.B. (2003) Early juvenile bioenergetic differences between anadromous and resident brook trout (*Salvelinus fontinalis*). *Canadian Journal of Aquatic Science*, **60**, 401–410.
- Morinville, G.R. & Rasmussen, J.B. (2008) Distinguishing between juvenile anadromous and resident brook trout (*Salvelinus fontinalis*) using morphology. *Environmental Biology of Fishes*, **81**, 171–184.
- Mueller, P. & Diamond, J. (2001) Metabolic rate and environmental productivity: well provisioned animals evolved to run and idle fast. *Proceedings of the National Academy of Sciences of the United States of America*, **98**, 12550–12554.
- Nicieza, N.A., Reiriz, L. & Brana, F. (1994) Variation in digestive performance between geographically disjunct populations of Atlantic salmon: countergradient in passage time and digestion rate. *Oecologia*, **99**, 243–251.
- Nislow, K.H., Folt, C.L. & Parrish, D.L. (1999) Favorable foraging locations for young Atlantic salmon: application to habitat and population restoration. *Ecological Applications*, **9**, 1085–1099.
- Petersson, E., Jarvi, T., Steffner, N.G. & Ragnarsson, B. (1996) The effect of domestication on some life history traits of sea trout and Atlantic salmon. *Journal of Fish Biology*, **48**, 776–791.
- Plaut, I. (2001) Critical swimming speed: its ecological relevance. *Comparative Biochemistry and Physiology*, **131**, 41–50.
- Post, J.R. & Evans, D.O. (1989) Size dependent over-winter mortality of young-of-the-year yellow perch (*Perca flavescens*): laboratory, *in situ* enclosure, and field experiments. *Canadian Journal of Fisheries and Aquatic Sciences*, **46**, 1958–1968.
- Post, J.R. & Parkinson, E.A. (2001) Energy allocation strategy in young fish: allometry and survival. *Ecology*, **82**, 1040–1051.
- Puckett, K.J. & Dill, L.M. (1985) The energetics of feeding territoriality in juvenile coho salmon (*Oncorhynchus kisutch*). *Behaviour*, **92**, 97–111.
- Quinn, T.P. & Peterson, N.P. (1996) The influence of habitat complexity and fish size on overwintering survival and growth of individually marked juvenile coho salmon (*Oncorhynchus kisutch*) in Big Beef Creek, Washington. *Canadian Journal of Fisheries and Aquatic Sciences*, **53**, 1555–1564.
- Ricker, W.E. (1975) Computation and interpretation of biological statistics of fish populations. *Bulletin of the Fisheries Research Board of Canada*, **191**.
- Ridgway, M.S. (2008) A roadmap for coasters: landscapes, life histories, and the conservation of brook trout. *Transactions of the American Fisheries Society*, **137**, 1179–1191.
- Robinson, B.W. & Wilson, D.S. (1994) Character release and displacement in fishes: a neglected literature. *The American Naturalist*, **144**, 596–627.
- Schluter, D. (1993) Adaptive radiation in sticklebacks: size, shape, and habitat use efficiency. *Ecology*, **74**, 699–709.
- Schluter, D. (1995) Adaptive radiation in sticklebacks: trade-offs in breeding performance and growth. *Ecology*, **76**, 82–90.
- Schultz, E.T. & Conover, D.O. (1997) Latitudinal differences in somatic energy storage: adaptive responses to seasonality in an estuarine fish (*Atherinidae: Menidia menidia*). *Oecologia*, **109**, 516–529.
- Shuter, B.J. & Post, J.R. (1990) Climate, population viability, and the zoogeography of temperate fishes. *Transactions of the American Fisheries Society*, **119**, 314–336.
- Sibly, R.M. (1981) Strategies of digestion and defecation. *Physiological Ecology: An Evolutionary Approach to Resource Use* (eds C.R. Townsend & P. Calow), pp. 109–139. Blackwell, Oxford.

- Speakman, J.R., Krol, E. & Johnson, M.S. (2004) The functional significance of individual variation in basal metabolic rate. *Physiological and Biochemical Zoology*, **77**, 900–915.
- Steffensen, J.F. (1989) Some errors in respirometry of aquatic breathers: how to avoid and correct for them. *Fish Physiology and Biochemistry*, **6**, 49–59.
- Stevens, D.E. (1992) Use of plastics materials in oxygen-measuring systems. *Journal of Applied Physiology*, **72**, 801–804.
- Steyermark, A.C. (2002) A high standard metabolic rate constrains growth. *Zoology*, **105**, 147–151.
- Sullivan, K., Martin, D.J., Cardwell, R.D., Toll, J.E. & Duke, S. (2001) *An Analysis of the Effects of Temperature on Salmonids of the Pacific Northwest with Implications for Selecting Temperature Criteria*. Sustainable Ecosystems Institute, Portland, OR.
- Sundstrom, F.L., Lohmus, M. & Devlin, R.H. (2005) Selection on increased intrinsic growth rates in coho salmon, *Oncorhynchus kisutch*. *Evolution*, **59**, 1560–1569.
- Taylor, E.B. & McPhail, J.D. (1985) Variation in body morphology among British Columbia populations of coho salmon, *Oncorhynchus kisutch*. *Canadian Journal of Fisheries and Aquatic Sciences*, **42**, 2020–2028.
- Tucker, S. & Rasmussen, J.B. (1999) Using ^{13}C s to measure and compare bioenergetic budgets of juvenile Atlantic salmon (*Salmo salar*) and brook trout (*Salvelinus fontinalis*) in the field. *Canadian Journal of Fisheries and Aquatic Sciences*, **56**, 875–887.
- Van Leeuwen, T.E. (2010) *Variation in Metabolic Rate Between Individuals and Species: Cryptic Physiological Tradeoffs Underlying Habitat Partitioning and Life History Strategies of Juvenile Salmonids*. MSc Thesis. University of British Columbia, Canada.
- Wieser, W. (1985) Developmental and metabolic constraints of the scope for activity in young rainbow trout (*Salmo gairdneri*). *Journal of Experimental Biology*, **118**, 133–142.
- Wurtsbaugh, W.A. & Davis, G.E. (1977) Effects of temperature and ration level on the growth and food conversion efficiency of *Salmo gairdneri*, Richardson. *Journal of Fish Biology*, **11**, 87–98.
- Young, K.A. (2001) *The Effects of Size Asymmetries and Habitat Diversity on Interspecific Interactions in Juvenile Salmonids (Oncorhynchus sp.)*. PhD Thesis. University of British Columbia, Canada.
- Young, K.A. (2004) Asymmetric competition habitat selection, and niche overlap in juvenile salmonids. *Ecology*, **85**, 134–149.

Received 21 September 2010; accepted 10 March 2011

Handling Editor: Bror Jonsson